

Olympiad Test Paper — Mathematics (Class 7)

Chapter 6: Number Play

Subject	Mathematics (NCERT Class 7)
Chapter	6 — Number Play
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This chapter rewards reasoning over guessing — use parity (odd/even) to rule out impossibilities, the "add the two previous terms" rule for the Virahāṅka–Fibonacci sequence, and careful place-value logic for digit puzzles. Watch the classic traps: an ODD count of odd numbers gives an ODD sum (not even); the sequence adds the two previous terms (not a constant); and in the "taller children in front" game the first child always says 0. Do not write on this paper — mark answers on your answer sheet.

1. In the "taller children in front" game, each child calls out how many TALLER children stand ahead of them in the line. Which of these numbers can the FIRST child in the line never say anything but?

- (A) the height of the tallest child (B) the number of children behind them
(C) any number from 0 upward (D) 0

2. A line of 5 children plays the "taller in front" game. Which of these called-out sequences (front to back) is IMPOSSIBLE?

- (A) 0, 1, 2, 3, 4 (B) 0, 1, 2, 3, 5
(C) 0, 0, 0, 0, 0 (D) 0, 1, 0, 2, 1

3. Five number cards, each showing an odd number, are laid down. Which total can they NEVER add up to?

- (A) 30 (B) 41
(C) 25 (D) 15

4. What is the parity of the sum $7 + 9 + 11 + 13 + 15 + 17$ (six odd numbers)?

- (A) it depends on the actual values (B) odd
(C) even (D) neither even nor odd

5. The Virahāṅka–Fibonacci sequence begins 1, 2, 3, 5, 8, 13, 21, What is the very next term after 21?

- (A) 33 (B) 42
(C) 34 (D) 29

6. In a cryptarithm, $T + T + T = UT$, where UT is a two-digit number whose units digit equals T . What digit is T ?

- (A) 9 (B) 0
(C) 2 (D) 5

7. A 3×3 grid is filled with the digits 1 to 9, each used exactly once. What is the SMALLEST sum any single row can possibly have?

- (A) 24 (B) 6
(C) 1 (D) 15

8. In a 3×3 grid filled with 1–9 (no repeats), the three row sums are 13, 17 and one unknown value. What is the third row sum?

- (A) 15 (B) 16
(C) 30 (D) 14

9. Ahaan claims he wrote down 7 whole numbers that add to an EVEN total, and that exactly 3 of them are odd. Is this possible?

- (A) Yes, if the even numbers are large enough
(B) No — an odd count of odd numbers makes the total odd
(C) Yes, if one of the numbers is zero (D) Only if all 7 numbers are odd

10. How many different rhythms of exactly 4 beats can be made using short (1-beat) and long (2-beat) syllables?

- (A) 4 (B) 6
(C) 8 (D) 5

11. The sum of three distinct digits chosen from 1–9 is 24. Which three digits must they be?

- (A) 7, 8, 9 (B) 8, 9, 6
(C) 9, 9, 6 (D) 6, 8, 7

12. Which of the following sums is ODD?

- (A) the sum of four odd numbers (B) the sum of the first 10 even numbers
(C) the sum of seven odd numbers
(D) the sum of three even numbers and two odd numbers

13. In the Virahāṅka–Fibonacci sequence 1, 1, 2, 3, 5, 8, 13, ... , what is the 9th term?

- (A) 21 (B) 55
(C) 13 (D) 34

14. A 3×3 grid is to be filled with 1–9 (no repeats). One row is required to sum to 26. Why is this impossible?

- (A) 26 is even and rows must be odd
- (B) The greatest possible row sum is $7 + 8 + 9 = 24$
- (C) 26 is not divisible by 3
- (D) Only the middle row can reach 26

15. Two odd numbers are written as $2m + 1$ and $2n + 1$. Their sum equals $2(m + n + 1)$. What does this PROVE?

- (A) odd + odd is sometimes even
- (B) the sum depends on m and n
- (C) odd + odd is always even
- (D) odd + odd is always odd

16. In the cryptarithm $T + T + T = UT$ (units digit of the answer equals T), what is the digit U ?

- (A) 1
- (B) 2
- (C) 3
- (D) 5

17. Which list of numbers follows the Virahāṅka–Fibonacci RULE (each term is the sum of the previous two), starting from 4 and 7?

- (A) 4, 7, 11, 18, 29
- (B) 4, 7, 10, 13, 16
- (C) 4, 7, 28, 196
- (D) 4, 7, 11, 19, 30

18. A child in the "taller in front" game says "3". What does this tell you for certain?

- (A) the child is the third tallest
- (B) there are exactly 3 children behind this child
- (C) the child is 3rd in line
- (D) exactly 3 taller children stand ahead of this child

19. Each consecutive pair of terms in 1, 1, 2, 3, 5, 8, 13, 21, ... has a ratio. As we go further along, these ratios get closer and closer to which value?

- (A) exactly 2
- (B) exactly 1.5
- (C) about 1.618
- (D) about 3.14

20. In the cryptarithm $AB + BA = 132$, where A and B are single digits, what is $A + B$?

- (A) 11
- (B) 12
- (C) 6
- (D) 13

21. Kishor wants to pick some odd cards that add to 30. Which number of odd cards CAN give a total of 30?

- (A) 4
- (B) 3
- (C) 5
- (D) 7

22. Using the identity $1 + 1 + 2 + 3 + 5 + \dots + F_n = F_{n+2} - 1$, find $1 + 1 + 2 + 3 + 5 + 8 + 13 + 21$.

- (A) 53 (B) 54
(C) 55 (D) 88

23. A 3×3 grid is filled with 1–9 (no repeats). What is the sum of ALL three row sums added together?

- (A) 36 (B) 40
(C) 30 (D) 45

24. In the "taller children in front" game with several children, which statement is ALWAYS true?

- (A) someone always says a number bigger than 0
(B) the called-out numbers are always in increasing order
(C) the first child always says 0 (D) every child says a different number

25. A two-digit number is a "palindrome" if reading it left-to-right gives the same as right-to-left (like 44 or 77). How many two-digit palindromes are there?

- (A) 10 (B) 18
(C) 9 (D) 11

26. Which of these statements about parity is the only one that is FALSE?

- (A) even + even = even (B) odd + odd = even
(C) even + odd = odd (D) odd + odd = odd

27. The Virahāṅka–Fibonacci sequence 1, 2, 3, 5, 8, What is the sum of its 6th and 7th terms (counting 1 as the 1st)?

- (A) 34 (B) 21
(C) 13 (D) 55

28. In a 3×3 grid of 1–9 (no repeats), one circled row sum is given as 4. Why is the grid impossible?

- (A) 4 is even
(B) the least sum of three distinct digits from 1–9 is 6
(C) the row repeats a digit (D) 4 is not a multiple of 3

29. The number 121 is a palindrome. What is the NEXT palindrome larger than 121?

- (A) 122 (B) 211
(C) 131 (D) 212

30. Three consecutive Virahāṅka–Fibonacci terms are x , y and 21, where $x + y = 21$. If $y = 13$, what is x ?

- (A) 5 (B) 8
(C) 9 (D) 7

31. A digit T satisfies $T + T =$ a two-digit number whose tens digit is 1. What is the smallest such T ?

- (A) 6 (B) 7
(C) 4 (D) 5

32. In a line of children playing the "taller in front" game, the last (back) child says "o". What does this mean?

- (A) no taller child stands ahead of the back child
(B) the line has only one child (C) the back child is the shortest
(D) the back child is shorter than everyone ahead

33. Which of the following can be the sum of FIVE consecutive whole numbers?

- (A) 32 (B) 36
(C) 38 (D) 30

34. In the cryptarithm $1A + A1 = 33$ (A is a single digit), what is A?

- (A) 1 (B) 2
(C) 3 (D) 0

35. How many different rhythms of exactly 5 beats can be made from short (1-beat) and long (2-beat) syllables?

- (A) 5 (B) 13
(C) 8 (D) 21

36. A sum of several odd numbers turned out to be even. Which is the only possible number of odd numbers below?

- (A) 6 (B) 5
(C) 3 (D) 1

37. The product (not sum) of two odd numbers is always:

- (A) odd (B) even
(C) a multiple of 4 (D) sometimes even, sometimes odd

38. In a 3×3 grid filled with 1–9, the first two columns sum to 15 and 18. What does the third column sum to?

- (A) 14 (B) 11
(C) 12 (D) 13

39. The first child in a "taller in front" line of 4 says "o". The numbers called are o, a, b, c. What is the GREATEST value c (the back child) could possibly say?

- (A) 4 (B) 3
(C) 2 (D) 1

40. Which number is divisible by 9?

- (A) 343 (B) 512
(C) 700 (D) 234

41. Two consecutive terms of a Virahāṅka–Fibonacci-type sequence are 19 and 31. What was the term just BEFORE the 19?

- (A) 12
- (B) 11
- (C) 13
- (D) 50

42. A teacher says: "I checked that $3 + 5 = 8$ is even, $7 + 9 = 16$ is even, and $11 + 13 = 24$ is even, so odd + odd is always even." What is the status of her reasoning?

- (A) It is a complete proof because she tried several cases
- (B) Examples can prove a statement if there are at least three
- (C) The conclusion is false
- (D) The conclusion is true but examples alone do not prove it

43. Starting a Virahāṅka–Fibonacci-type rule with 2 and 5 gives 2, 5, 7, 12, 19, What is the term after 19?

- (A) 24
- (B) 26
- (C) 31
- (D) 38

44. In a cryptarithm each DIFFERENT letter must stand for a DIFFERENT digit. In " $AB + BA = 132$ ", why can A and B not both equal 6?

- (A) because $66 + 66 = 132$ and that would be allowed
- (B) because different letters must be different digits
- (C) because 6 is not a valid digit
- (D) because A must always be larger than B

45. How many odd numbers are there from 1 to 20 inclusive, and is their SUM odd or even?

- (A) 10 odd numbers, sum even
- (B) 10 odd numbers, sum odd
- (C) 9 odd numbers, sum odd
- (D) 11 odd numbers, sum even

46. A 3×3 grid is filled with 1–9 (no repeats). The top row sums to 6 and the bottom row sums to 24. What must the middle row sum to?

- (A) 14
- (B) 16
- (C) 15
- (D) 12

47. The Virahāṅka–Fibonacci sequence (1, 2, 3, 5, 8, 13, ...) has terms that are alternately odd and even in the pattern O, E, O, O, E, O, O, E, What is the parity of the 8th term (counting from 1 as the 1st)?

- (A) odd
- (B) even
- (C) it alternates each time
- (D) cannot be determined

48. Which group of three distinct digits from 1–9 has a sum that is NOT possible as a single row of a 3×3 grid of 1–9?

- (A) 1, 2, 3
- (B) 7, 8, 9
- (C) 4, 5, 6
- (D) 1, 2, 2

49. When a whole number N is doubled, the result always ends in an EVEN digit. Which of these can therefore NEVER be the units digit of $2N$?

- (A) 7 (B) 4
(C) 0 (D) 8

50. In the cryptarithm $AA + A = BA$, where A and B are single non-zero digits and the answer's units digit is A , what is A ?

- (A) 9 (B) 5
(C) impossible for any such A (D) 7

51. The sum $2 + 4 + 6 + \dots + 20$ (the first ten even numbers) is:

- (A) 100 (B) 90
(C) 120 (D) 110

52. A palindrome reads the same forwards and backwards. Which of these three-digit numbers is a palindrome?

- (A) 120 (B) 252
(C) 102 (D) 210

53. Reading the "taller in front" sequence 0, 1, 1, 3 for a line of 4 children: which statement MUST be false?

- (A) the 2nd child has 1 taller child ahead (B) the 4th child has 3 taller children ahead
(C) the 1st child says a number larger than 0 (D) the 3rd child has 1 taller child ahead

54. The 10th and 11th terms of 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ... are 55 and 89. What is the 12th term?

- (A) 121 (B) 134
(C) 110 (D) 144

55. Kishor adds an EVEN count of odd numbers. The total is guaranteed to be:

- (A) odd (B) even
(C) a multiple of 4 (D) sometimes odd, sometimes even

56. In the cryptarithm $ABC \times 1 = CBA$, where ABC is a three-digit number, which kind of number must ABC be?

- (A) a palindrome (B) a multiple of 9
(C) an even number (D) a multiple of 11 only

57. A number when written and then reversed gives the same value, and it lies between 100 and 200. How many such numbers are there?

- (A) 11 (B) 9
(C) 10 (D) 20

58. Six children play "taller in front" and call out 0, 1, 2, 1, 4, 0. Which child's number is DEFINITELY impossible given that only 6 children are in the line?

- (A) the child who said 4 (B) the 1st child who said 0
(C) the last child who said 0
(D) none — all six numbers are individually possible

59. Using the rule that the sum of the first n Virahāṅka–Fibonacci terms (1, 1, 2, 3, 5, 8) equals $F_{n+2} - 1$, find $1 + 1 + 2 + 3 + 5$.

- (A) 12 (B) 11
(C) 13 (D) 15

60. Five distinct odd numbers are added. Without knowing the numbers, which of these can you be SURE the total is NOT?

- (A) 35 (B) 100
(C) 49 (D) 63

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Mathematics (Class 7), Chapter 6: Number Play

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	A	21	A	31	D	41	A	51	D
2	B	12	C	22	B	32	A	42	D	52	B
3	A	13	D	23	D	33	D	43	C	53	C
4	C	14	B	24	C	34	B	44	B	54	D
5	C	15	C	25	C	35	C	45	A	55	B
6	D	16	A	26	D	36	A	46	C	56	A
7	B	17	A	27	A	37	A	47	B	57	C
8	A	18	D	28	B	38	C	48	D	58	D
9	B	19	C	29	C	39	B	49	A	59	A
10	D	20	B	30	B	40	D	50	C	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) The first child in the line has nobody standing ahead of them, so the number of taller children in front is exactly 0 — always. (Being the "tallest" is option-A's trap: saying 0 only means no one taller is ahead, which is automatic for the front position regardless of height.)

2. (B) A child can name at most as many taller children as there are people ahead. In a line of 5, the back child has only 4 ahead, so the largest possible call is 4. "0, 1, 2, 3, 5" ends in 5 — impossible. The others obey the bound (every value $\leq$ number ahead).

3. (A) Five odd numbers always add to an ODD total (odd count of odds $\rightarrow$ odd). 30 is even, so it can never be the sum. 41, 25 and 15 are all odd and are achievable. So the impossible total is 30.

4. (C) Six odd numbers is an EVEN count of odds, so the sum is even. (Actual sum $7+9+11+13+15+17 = 72$, even.) Parity is fixed by the count, not the values — so "it depends" is the trap.

5. (C) Each term is the sum of the two before: after ..., 13, 21 the next is $13 + 21 = 34$. (Adding a constant like +12 to get 33 is the "fixed gap" misconception.)

6. (D) $T + T + T = 3T$ must be a two-digit number ending in T. Try $T = 5$: $3 \times 5 = 15$, ends in 5 = T, with U = 1. ✓ $T = 2$ gives 6 (one digit, no U); larger T overshoots the ending-in-T condition. So $T = 5$.

7. (B) The smallest a row can be is the three smallest distinct digits: $1 + 2 + 3 = 6$. (1 would need a single cell, not a row sum — that's the trap.)

8. (A) The three row sums always total $1+2+\dots+9 = 45$ (the 45-invariant). So the third = $45 - 13 - 17 = 15$.

9. (B) Exactly 3 odd numbers is an ODD count of odds, which forces an ODD total. An odd total can never be even, so it is impossible — no matter how large the even numbers are. Answer: No.

10. (D) Rhythm counts follow 1, 2, 3, 5, ... (each the sum of the previous two). For 1 beat: 1 way; 2 beats: 2; 3 beats: 3; 4 beats: $3 + 2 = 5$. So 5 rhythms.

11. (A) Three distinct digits from 1–9 summing to 24 must be the three largest, $7 + 8 + 9 = 24$ (the maximum). $8+9+6 = 23$ and $6+8+7 = 21$ fall short; 9,9,6 repeats a digit. So 7, 8, 9.

12. (C) Parity of each: four odds $\rightarrow$ even; first 10 evens $\rightarrow$ even; three evens + two odds $\rightarrow$ even (two odds make even); seven odds $\rightarrow$ ODD (odd count of odds). So the odd one is "the sum of seven odd numbers."

13. (D) 1, 1, 2, 3, 5, 8, 13, 21, 34 — the 9th term is 34. (21 is the 8th — an off-by-one trap.)

14. (B) The largest a row of three distinct digits from 1–9 can reach is $7 + 8 + 9 = 24$. Since $26 > 24$, no row can sum to 26. (Divisibility by 3 is irrelevant here.)

15. (C) $(2m+1) + (2n+1) = 2m + 2n + 2 = 2(m + n + 1)$, a multiple of 2 — hence even — for ALL whole m, n. This is a general proof that odd + odd is ALWAYS even (not just in examples).

16. (A) From Q6, $5 + 5 + 5 = 15$, so UT = 15: the tens digit U = 1.

17. (A) Apply "add the previous two" from 4, 7: $4+7 = 11$, $7+11 = 18$, $11+18 = 29 \rightarrow 4, 7, 11, 18, 29$. (Option B adds a constant 3; D mis-adds $11+8$.)

18. (D) The number a child calls is, by the rule of the game, exactly how many TALLER children stand ahead of them — here, 3. It says nothing certain about their rank, position in line, or how many are behind.

19. (C) Ratios of consecutive terms ($2/1, 3/2, 5/3, 8/5, 13/8, \dots$) close in on the golden ratio $\phi \approx 1.618$. (Not 2, 1.5, or $\pi \approx 3.14$.)

20. (B) $AB + BA = (10A + B) + (10B + A) = 11A + 11B = 11(A + B) = 132 \rightarrow A + B = 132 \div 11 = 12$.

21. (A) An EVEN count of odd cards gives an even sum; an odd count gives odd. 30 is even, so the count must be even — among the options only 4 is even. (4 odds, e.g. $3+5+9+13 = 30$. ✓)

22. (B) Add directly: $1+1 = 2, +2 = 4, +3 = 7, +5 = 12, +8 = 20, +13 = 33, +21 = 54$. (The identity gives $F_{10} - 1 = 55 - 1 = 54$ too — choosing 55 forgets the "-1".)

23. (D) The nine entries are 1–9 once each, totalling 45; the three row sums partition all nine cells, so they add to 45 (the 45-invariant) regardless of arrangement.

24. (C) The front child has no one ahead, so they always call o — always true. The numbers need not increase, need not all differ, and no one is forced to exceed o (e.g. everyone in increasing-height order calls o).

25. (C) Two-digit palindromes have equal digits: 11, 22, 33, 44, 55, 66, 77, 88, 99 — that's 9. (10 wrongly counts 00; 11/18 over-count.)

26. (D) even+even = even ✓, odd+odd = even ✓, even+odd = odd ✓ are all true. "odd + odd = odd" is FALSE (two odds give an even). So the false one is (D).

27. (A) Sequence 1, 2, 3, 5, 8, 13, 21: 6th term = 13, 7th term = 21, sum = $13 + 21 = 34$. (This sum is also the 8th term.)

28. (B) The smallest possible sum of three distinct digits from 1–9 is $1 + 2 + 3 = 6$, so a row can never sum to 4. (Being even or not a multiple of 3 is not the reason.)

29. (C) Palindromes just above 121: 122 isn't a palindrome (221 reversed), but 131 reads the same both ways and is the next one larger than 121. (211 reverses to 112; 212 is larger than 131.)

30. (B) Consecutive terms satisfy $x + y = (\text{next term}) = 21$. With $y = 13$, $x = 21 - 13 = 8$. (Indeed ..., 8, 13, 21, ... is the Fibonacci run.)

31. (D) $2T$ must be a two-digit number with tens digit 1, i.e. $10 \leq 2T \leq 19$. The smallest T with $2T \geq 10$ is $T = 5$ ($2 \times 5 = 10$). ($T = 4$ gives 8, only one digit.)

32. (A) Calling "o" always means no taller child stands ahead — that is its exact meaning for any position, including the back. It does NOT force the line to have one child or make that child the shortest (others ahead could be shorter).

33. (D) Five consecutive whole numbers $n, \dots, n+4$ sum to $5n + 10 = 5(n + 2)$, a multiple of 5. Among the options only 30 is a multiple of 5 ($30 = 4+5+6+7+8$). 32, 36, 38 are impossible.

34. (B) $1A + A1 = (10 + A) + (10A + 1) = 11 + 11A = 33 \rightarrow 11A = 22 \rightarrow A = 2$. Check: $12 + 21 = 33$. ✓

35. (C) Rhythm counts: 1, 2, 3, 5, 8 for 1, 2, 3, 4, 5 beats. So 5 beats give 8 rhythms ($5 + 3$ from the recurrence). (13 is for 6 beats.)

36. (A) A sum of odds is even only when the COUNT of odds is even. Among the options only 6 is even, so 6 odd numbers can sum to even. (5, 3, 1 are odd counts $\rightarrow$ odd total.)

37. (A) Odd $\times$ odd = odd: writing them $2m+1$ and $2n+1$, the product is $4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$, which is one more than an even number — odd. (Multiple of 4 fails, e.g. $3 \times 5 = 15$.)

38. (C) The three column sums also total 45 (every cell counted once). So the third column $= 45 - 15 - 18 = 12$.

39. (B) In a line of 4, the back child has 3 people ahead, so at most 3 of them can be taller — the greatest value c can call is 3. (4 would need 4 children ahead, but only 3 exist.)

40. (D) A number is divisible by 9 when its digit sum is. $2+3+4 = 9$ ✓ (234). $3+4+3 = 10$, $5+1+2 = 8$, $7+0+0 = 7$ — none divisible by 9. So 234.

41. (A) In an "add the previous two" sequence, the term before 19 plus 19 gives the next term 31, so it is $31 - 19 = 12$. Check: 12, 19, 31 ($12 + 19 = 31$). ✓

42. (D) Checking three examples cannot PROVE a universal claim — examples can only suggest a pattern or disprove it with a counterexample. The conclusion happens to be true (provable by $2m+1 + 2n+1 = \text{even}$), but her reasoning by examples is not a proof.

43. (C) Continue 2, 5, 7, 12, 19 by adding the last two: $12 + 19 = 31$. (Same recurrence, different start.)

44. (B) In a cryptarithm, different LETTERS must stand for different digits; A and B are different letters, so they cannot both be 6. (Also $66 + 66 = 132$ numerically, but the rule, not the arithmetic, forbids it.)

45. (A) Odd numbers from 1 to 20: 1, 3, 5, ..., 19 — that's 10 of them. Ten is an EVEN count of odds, so their sum is even (in fact 100). So: 10 odd numbers, sum even.

46. (C) Row sums total 45. With top = 6 and bottom = 24, the middle $= 45 - 6 - 24 = 15$.

47. (B) Terms 1, 2, 3, 5, 8, 13, 21, 34: the 8th term is 34, which is even. (The parity pattern O, E, O, O, E, O, O, E confirms the 8th slot is E.)

48. (D) A valid row uses three DISTINCT digits. "1, 2, 2" repeats 2, so it cannot be a row of a 1–9 grid (every digit appears once). $1+2+3 = 6$, $7+8+9 = 24$, $4+5+6 = 15$ are all legal sums; 1,2,2 is the impossible set.

49. (A) Doubling any whole number gives an even result, so $2N$ always ends in an even digit (0, 2, 4, 6 or 8). 7 is odd, so it can NEVER be the units digit of $2N$. (4, 0, 8 are all even and possible.)

50. (C) $AA + A = 11A + A = 12A$. For the answer to end in A we need $12A \equiv A \pmod{10}$, i.e. $2A \equiv 0 \pmod{10}$, so $A = 0$ or 5; A is non-zero, so $A = 5 \rightarrow 12 \times 5 = 60$, which ends in 0, not 5. No non-zero digit works, so it is impossible.

51. (D) $2 + 4 + \dots + 20 = 2(1 + 2 + \dots + 10) = 2 \times 55 = 110$. (90 forgets the doubling; 100 is the count-of-odds sum from Q45.)

52. (B) A palindrome reads the same reversed. 252 reversed is 252 ✓. $120 \rightarrow 021$, $102 \rightarrow 201$, $210 \rightarrow 012$ — none match. So 252.

53. (C) In the sequence 0, 1, 1, 3 the FIRST child calls 0 (nobody ahead), so "the 1st child says a number larger than 0" MUST be false. The other statements correctly read off the sequence (2nd $\rightarrow$ 1, 4th $\rightarrow$ 3, 3rd $\rightarrow$ 1), all of which are possible.

54. (D) 12th term = 10th + 11th = $55 + 89 = 144$. (Adding wrong pairs gives the traps 134/110/121.)

55. (B) An EVEN count of odd numbers always sums to an even number (the odds pair up). It need not be a multiple of 4 (e.g. $3 + 3 = 6$). So the total is guaranteed even.

56. (A) $ABC \times 1 = ABC$, and the puzzle sets this equal to CBA. $ABC = CBA$ means the number reads the same reversed — it must be a palindrome. (It need not be even, a multiple of 9, or a multiple of 11.)

57. (C) A number between 100 and 200 reading the same reversed has the form 1X1 (first and last digit 1). X can be 0–9: 101, 111, ..., 191 — that's 10 numbers.

58. (D) Each value must not exceed the number of children ahead: positions 1–6 have 0,1,2,3,4,5 ahead. The calls 0,1,2,1,4,0 satisfy $0 \leq 0$, $1 \leq 1$, $2 \leq 2$, $1 \leq 3$, $4 \leq 4$, $0 \leq 5$ — every value is individually possible. So none is definitely impossible.

59. (A) Add directly: $1 + 1 + 2 + 3 + 5 = 12$. (The identity gives $F_7 - 1 = 13 - 1 = 12$ too — choosing 13 forgets the "–1".)

60. (B) Five distinct odd numbers form an ODD count of odds, so their sum is always ODD. 100 is even, so the total can never be 100. (35, 49, 63 are all odd and achievable.)

Solutions complete: 60 / 60.